

Executive Summary: This briefing explains the architectural updates made to resolve account deletion security, studio closure workflows, and the artisan re-application flow (resolving the missing email issue and model imports). All changes preserve cross-module compatibility and pass all 114 automated test suites.

1. Key Architectural Changes

Option A: "Close Studio & Revert to Explorer"

For artisans retiring their business without deleting their personal identity:

- Role Demotion:** Role reset from `Artisan` to `Tourist` (Cultural Explorer).
- Studio Profile:** `artisan_profiles.status` set to `CLOSED`. Studio name, category, and SSM cleared from active profile.
- Quests Retired:** Active quests retired to prevent orphaned live quests.
- Tourist Data Retained:** All collected badges, stamps, favorites, and reviews are preserved.
- Real-time Directory Refresh:** Triggers `DirectoryViewModel.fetchArtisans()` immediately so the closed studio disappears from search without needing an app reboot.

Option B: "Delete Account Permanently"

For full account termination across all roles:

- Password Verification:** Requires active password confirmation via `reauthenticateWithPassword` to prevent unauthorized deletion.
- Clean Error Messaging:** Replaced raw technical string `AuthException(message: Invalid login credentials...)` with a clean, user-friendly alert: *"Incorrect password. Please verify your current password."*
- Complete Purge:** Wipes public profile, artisan records, submitted documents, and Supabase auth record.
- Identifier Release:** Releases username and email so they can be re-used in future registrations.

2. Artisan Re-Application Flow Fixes

Problem Identified: When an artisan closed their studio and subsequently attempted to apply again via `ApplyArtisanScreen`, the submission failed with:

Exception: User email is required to submit artisan application.
followed by compile error: 'UserModel' isn't a type in `apply_artisan_screen.dart:486`.

Root Causes & Solutions Implemented:

- Missing Import (`UserModel`):** Added `import 'package:warisan_kita/domain/models/user.dart';` to `apply_artisan_screen.dart` to resolve the compilation error.
- Empty Email on Re-Application:** If the user applied immediately after role demotion, the in-memory `authVM.currentUser` was either reloading or empty. We implemented a **3-tier resilient fallback**:
 - UI Layer (`ApplyArtisanScreen`):** If `currentUser` or email is empty, it calls `authVM.restoreSession()` prior to submission.
 - ViewModel Layer (`AuthViewModel`):** If passed email is empty, it resolves directly from session or active user profile.
 - Service Layer (`SupabaseService.linkArtisanRoleToTourist`):** Checks active Supabase auth session (`client.auth.currentUser?.email`), local persistent preferences (`_keyAuthEmail` & `_keyAuthUser`), and memory store before throwing errors.

3. Modified Files Summary

File	Component / Class	Description of Modification
lib/data/services/supabase_service.dart	deactivateArtisanStudio linkArtisanRoleToTourist	Demotes role to Tourist; closes studio; retires quests; clears studio metadata; implements 3-tier email fallback.
lib/viewmodels/auth_viewmodel.dart	linkArtisanToExistingTourist	Auto-resolves current user email from restored session if empty string is supplied.
lib/ui/tourist/apply_artisan_screen.dart	_submitApplication	Imports UserModel ; triggers session restoration fallback before sending artisan application.
lib/ui/artisan/artisan_settings_screen.dart	_confirmCloseStudio	Refreshes DirectoryViewModel on studio deactivation; displays user-friendly error banners.
lib/ui/core/settings_screen.dart	_showDeleteAccountDialog	Translates raw Supabase AuthExceptions into human-friendly password error messages.
supabase_schema.sql	deactivate_artisan_studio	Added Postgres stored procedure with SECURITY DEFINER for clean backend studio retirement.
test/authentication_test.dart	Authentication Test Suite	Added automated test for studio deactivation followed by seamless artisan re-application.
test/support/auth_backend.dart	Mock Backend Server	Added mock handling for quest retirement and artisan profile insertions.

4. Verification & Test Suite Results

All 114 Automated Tests Passed (100% Success Rate):			
Test Suite	Test Count	Status	Coverage Focus
authentication_test.dart	36 tests	PASSED	Login, password change, studio closure, account deletion, re-application
profile_validation_test.dart	27 tests	PASSED	SSM checks, document verification, geolocation moderation
Remaining Modules (11 test files)	51 tests	PASSED	Quests, maps, matchmaking, badges, chat, admin moderation
Total Test Suite	114 tests	100% PASS	Zero regressions across all mobile and web flows

5. Team Action Items

- Pull Latest Changes:** Run `git pull origin Tze-Chuen` to fetch the updated branch.
- Database Migration (Optional):** The SQL function `deactivate_artisan_studio` has been added to `supabase_schema.sql` for production deployments.

- **Run Tests:** Run `flutter test` to confirm all 114 tests pass on your machine.